BIRZEIT UNIVERSITY

Electrical and Computer Engineering Department
ENEE2313, Signals and Systems, Assignment

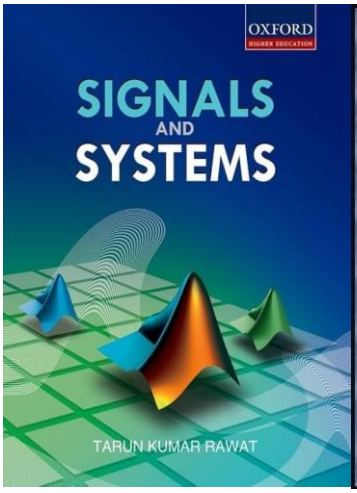
Matlab Assignment solution

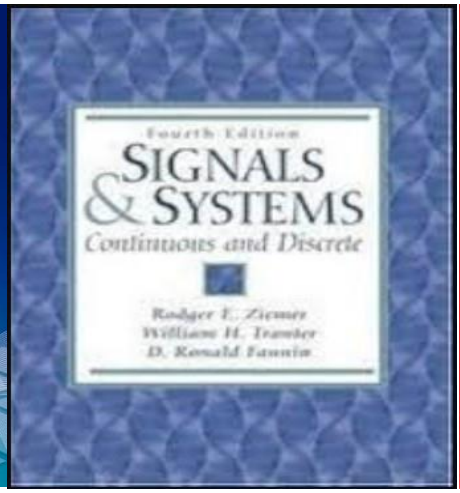
Name: Nagham Maali.

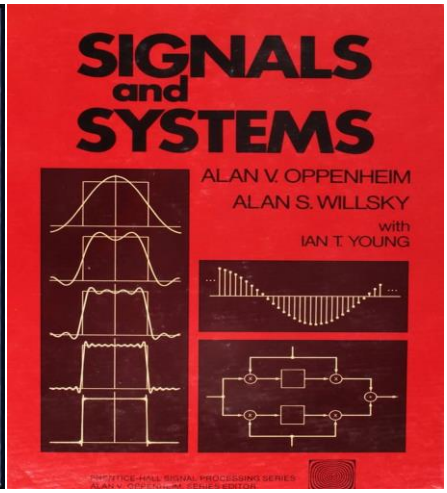
ID NO.: 1212312.

Section: 3.

Instructor: Dr. Ashraf Al–Rimawi.







<u>Problem 1:</u>

Generate and plot the following signals using MATLAB:

a. $x1\ (t) = 2\Pi\ (5-t\ /4) + \Pi\ (t+3\ /2) - r\ (t-12\ /2) + 2u\ (t - 16)$

- **To determine the time interval for the signal $x1$:**
  - ✓ **$\Pi\ (5-t\ /4)$:**
    -1/2 <= 5 – t /4 <= 1/2
    And so: 3 <= t <= 7

  - ✓ **$\Pi\ (t+3\ /2)$:**
    -1/2 <= t + 3 /2 <= 1/2
    And so: -4 <= t <= -2

  - ✓ **$r\ (t-12\ /2)$:**
    Will start at t = 12

  - ✓ **$u\ (t - 16)$:**
    Will start at t = 16

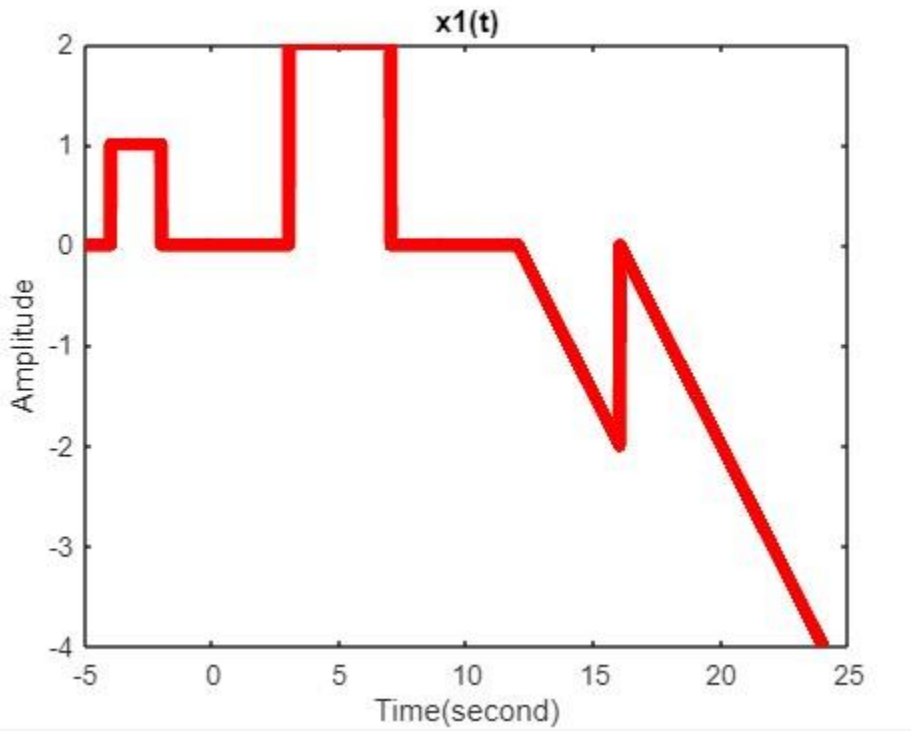
So the suitable interval:]-5 , 24]

- **Code:**

```
TimeInterval = -5:0.01:24 ;

x1 = (2 * rectpuls((5 - TimeInterval)/4)) +
rectpuls((TimeInterval + 3)/2) - (((TimeInterval - 12)/2) .*
(heaviside((TimeInterval - 12)/2))) + (2 *
heaviside(TimeInterval - 16)) ;

plot(TimeInterval , x1 , 'r' , 'LineWidth' , 5) ;
title('x1(t)') ;
xlabel('Time(second)') ;
```

```
ylabel('Amplitude') ;
```

- **Figure:**



b. $x2\ (t) = 2\Pi\ (t-4\ /6) - r\ (t-6\ /2) + r\ (-t + 6)$

- **To determine the time interval for the signal $x1$:**
  - ✓ **$\Pi\ (t-4\ /6)$:**
    -1/2 <= t - 4 /6 <= 1/2
    And so: 1 <= t <= 7

  - ✓ **$r\ (t-6\ /2)$:**
    Will start at t = 6

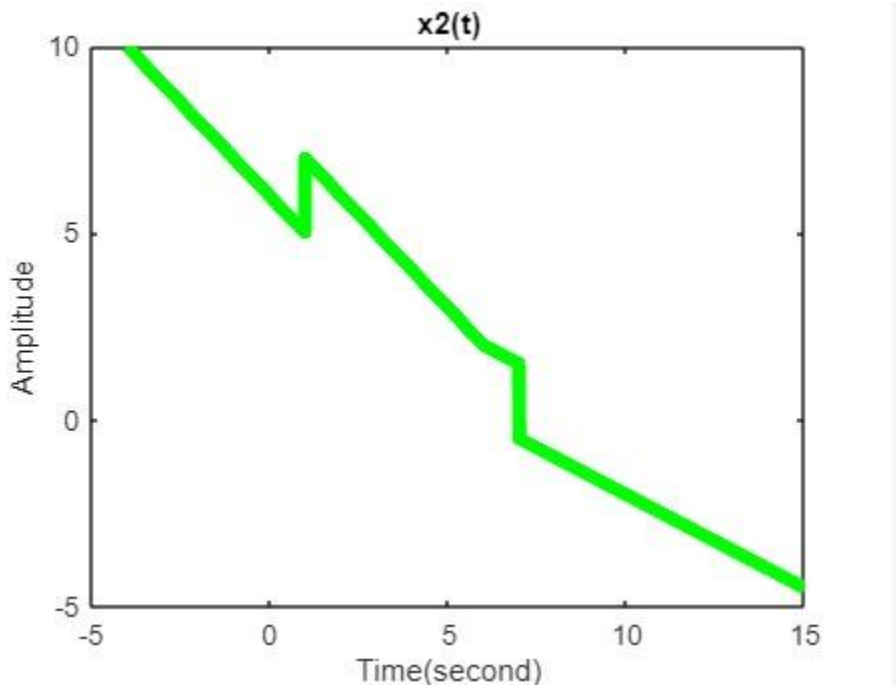
  - ✓ **$r\ (-t + 6)$:**
    t <= 6

So the suitable interval:]-4 , 15]

- **Code:**

```
TimeInterval = -4:0.01:15 ;

x2 = (2 * rectpuls((TimeInterval - 4)/6)) - (((TimeInterval - 6)/2)
.* (heaviside((TimeInterval - 6)/2))) + ((6 - TimeInterval) .*
(heaviside(6 - TimeInterval))) ;

plot(TimeInterval , x2 , 'g' , 'LineWidth' , 5) ;
title('x2(t)') ;
xlabel('Time(second)') ;
ylabel('Amplitude') ;
```

- **Figure:**



<u>Problem 2:</u>

Consider the following signals:

$x1\ (t) = 2\cos(20\pi t)$ and $x2\ (t) = 3\sin(60\pi t)$

A. Generate and plot $x1(t)$ for one period.

W = 20pi

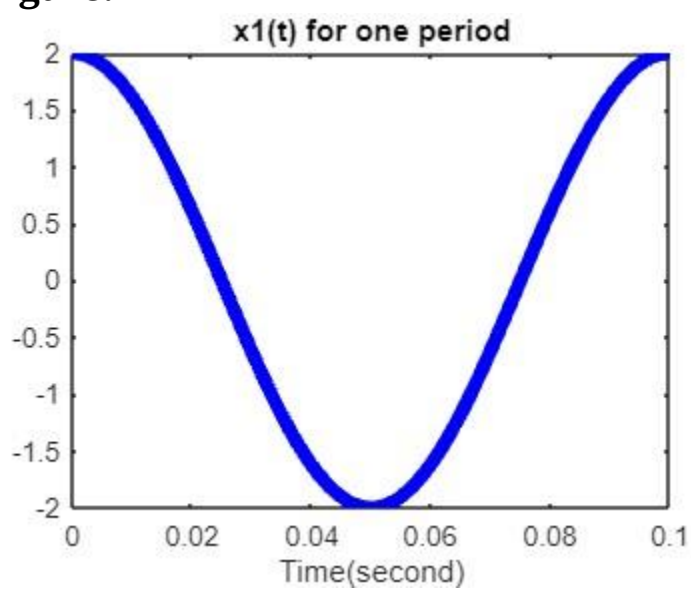
2pi/T = 20pi

T = 1/10 → period

- **Code:**

```
T = 1/10 ;  %one period
TimeInterval = 0:0.001:T ;

x1 = 2 * cos(20 * pi * TimeInterval) ;

plot(TimeInterval , x1 , 'b' , 'LineWidth' , 5) ;
title('x1(t) for one period') ;
xlabel('Time(second)') ;
```

- **Figure:**



B. Generate and plot $xb(t) = x1(t) + x2(t)$ for one period.

W1 = 20pi
2~~pi~~/T1 = 20~~pi~~
T1 = 1/10 → period of x1

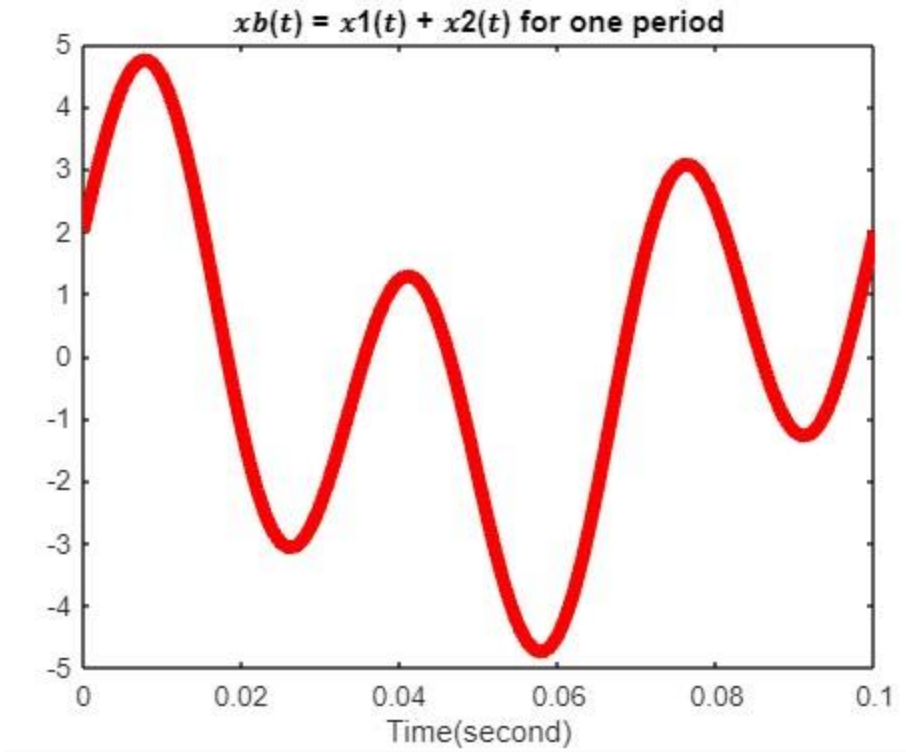
W2 = 60pi
2~~pi~~/T2 = 60~~pi~~
T2 = 1/30 → period of x2

Common period: LCM (T1, T2) = LCM (1, 1)/GCD (10, 30)
= 1/10

- **Code:**

```
T = 1/10 ;  %LCM period
TimeInterval = 0:0.001:T ;

x1 = 2 * cos(20 * pi * TimeInterval) ;
x2 = 3 * sin(60 * pi * TimeInterval) ;

xb = x1 + x2 ;

plot(TimeInterval , xb , 'r' , 'LineWidth' , 5) ;
title('xb(t) = x1(t) + x2(t) for one period') ;
xlabel('Time(second)') ;
```

- **Figure:**



C. Generate and plot $xc(t) = x1(t).\ x2(t)$ for one period.

W1 = 20pi

2pi/T1 = 20pi

T1 = 1/10 → period of $x1$

W2 = 60pi

2pi/T2 = 60pi

T2 = 1/30 → period of $x2$

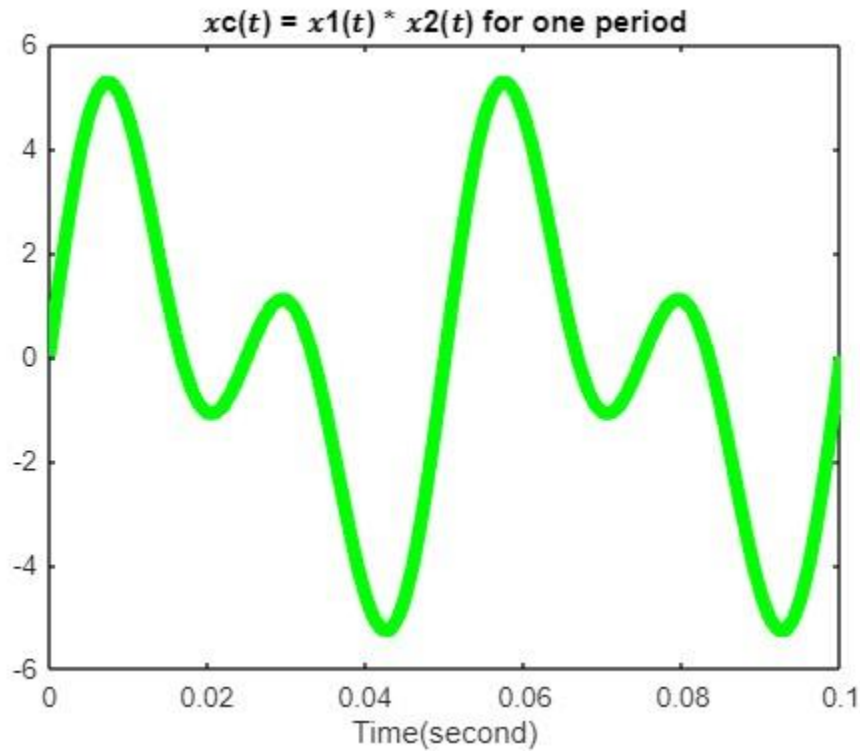
Common period: LCM (T1, T2) = LCM (1, 1)/GCD (10, 30)
= 1/10

- **Code:**

```
T = 1/10 ;  %LCM period
TimeInterval = 0:0.001:T ;

x1 = 2 * cos(20 * pi * TimeInterval) ;
```

```
x2 = 3 * sin(60 * pi * TimeInterval) ;

xc = x1 .* x2 ;

plot(TimeInterval , xc , 'g' , 'LineWidth', 5) ;
title('xc(t) = x1(t) * x2(t) for one period') ;
xlabel('Time(second)') ;
```

- **Figure:**

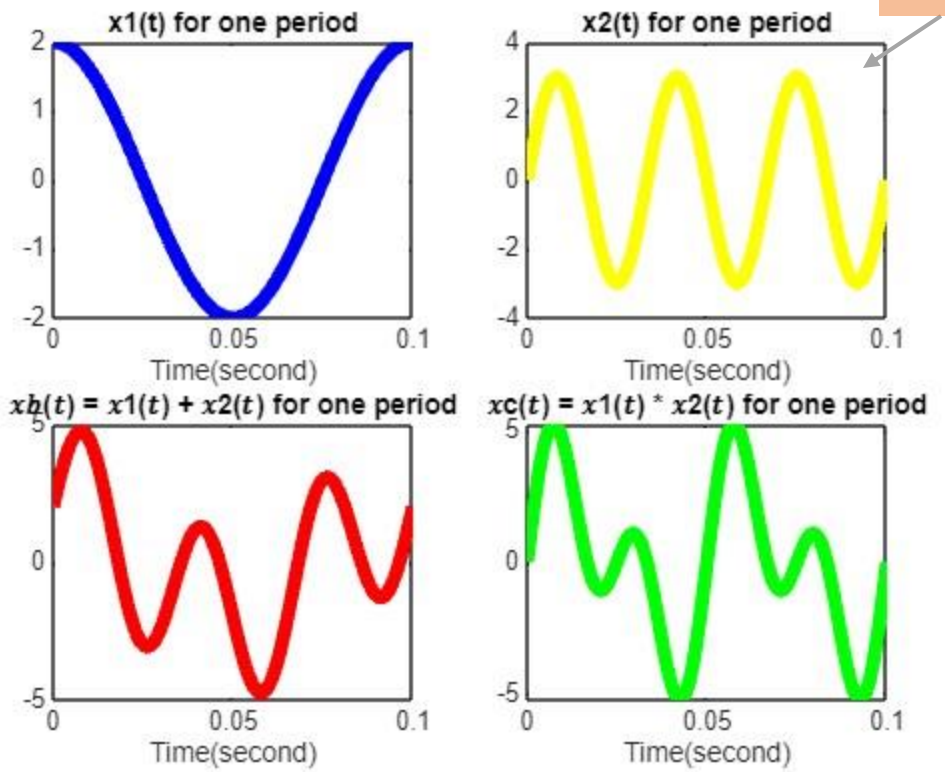


Show all the results on one the same figure using subplot.

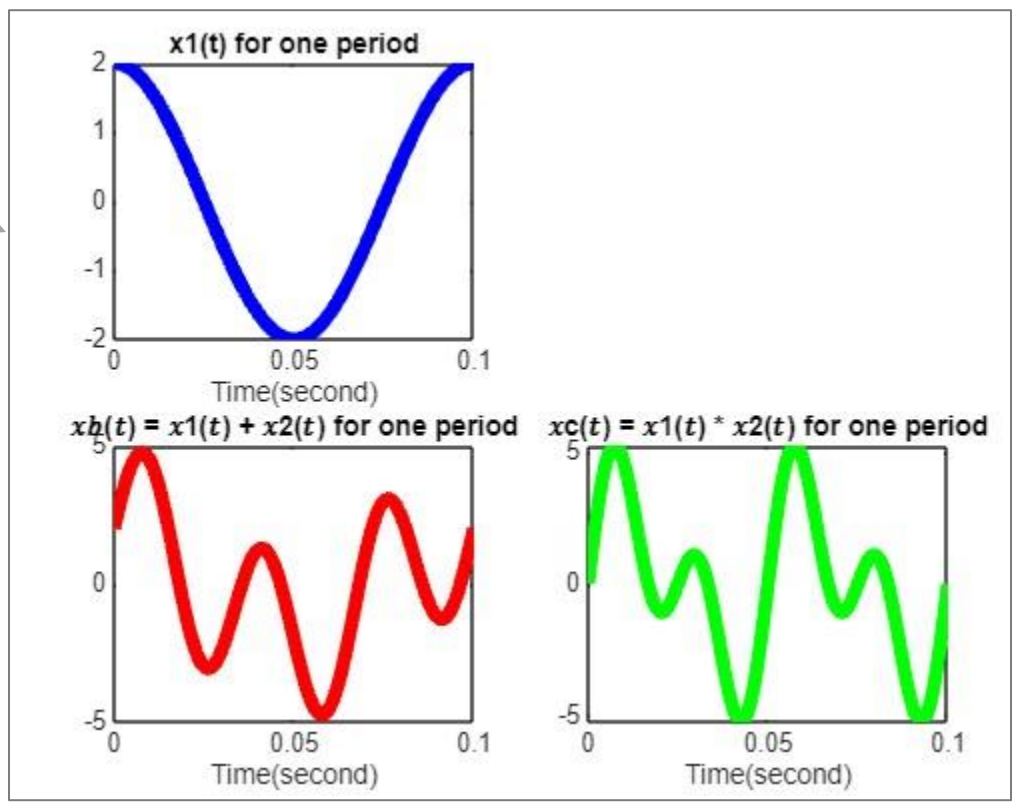
```
T = 1/10 ;  %LCM period
TimeInterval = 0:0.001:T ;

x1 = 2 * cos(20 * pi * TimeInterval) ;
x2 = 3 * sin(60 * pi * TimeInterval) ;
```

```
xb = x1 + x2 ;
xc = x1 .* x2 ;

subplot(2 , 2 , 1) ;
plot(TimeInterval , x1 , 'b' , 'LineWidth' , 5) ;
title('x1(t) for one period') ;
xlabel('Time(second)') ;

subplot(2 , 2 , 2) ;
plot(TimeInterval , x2 , 'y' , 'LineWidth' , 5) ;
title('x2(t) for one period') ;
xlabel('Time(second)') ;

subplot(2 , 2 , 3) ;
plot(TimeInterval , xb , 'r' , 'LineWidth' , 5) ;
title('xb(t) = x1(t) + x2(t) for one period') ;
xlabel('Time(second)') ;

subplot(2 , 2 , 4) ;
plot(TimeInterval , xc , 'g' , 'LineWidth' , 5) ;
title('xc(t) = x1(t) * x2(t) for one period') ;
xlabel('Time(second)') ;
```

Wrong period



Without x2(t)



<u>Problem 3:</u>

A periodic signal is defined by the following:

$$x(t) = 10\sin(\omega_0 t)\sum_{n=-\infty}^{\infty}\left[\Pi\left(\frac{t-\frac{3\,T_0}{4}-nT_0}{\frac{T_0}{4}}\right)+\Pi\left(\frac{t+\frac{3\,T_0}{4}+nT_0}{\frac{T_0}{4}}\right)\right]$$

Using input command to enter the value of *T*0, and plot the first two periods of the signal.

The first two periods: n = 0 and n = 1

And with reformation and unification of denominators ...

x(t) = 10 sin(w0t) (Π (((4 * t) – (3 * T0)) / T0) + Π (((4 * t) + (3 * T0)) / T0) + Π (((4 * t) – (7 * T0)) / T0) + Π (((4 * t) + (7 * T0)) / T0))

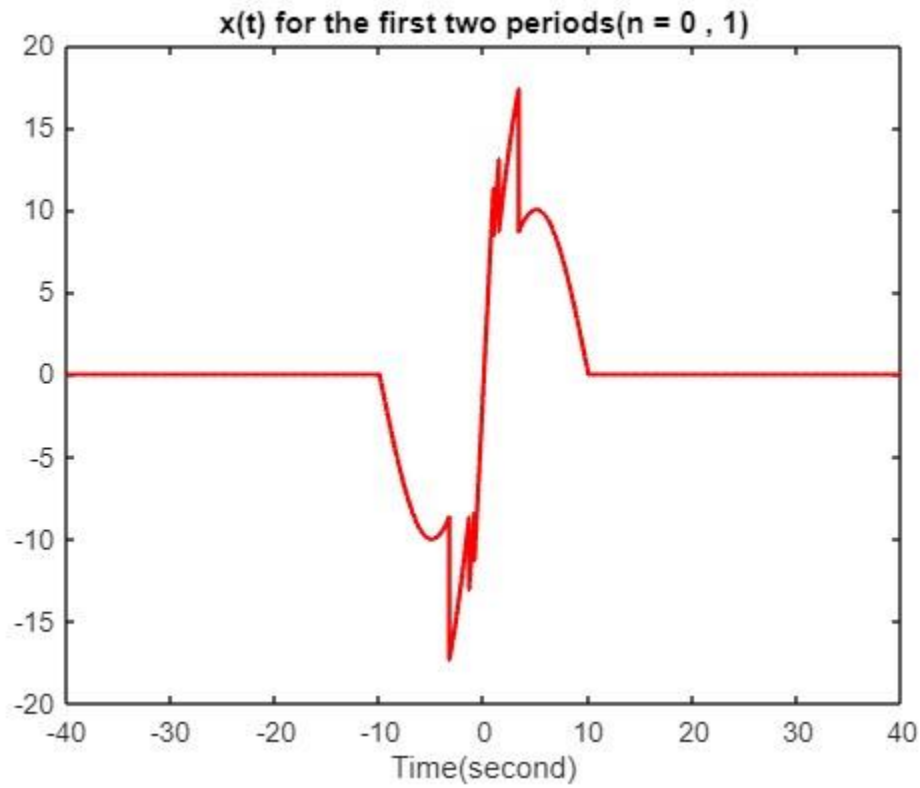
w0 = (2 * pi) / To

- **Code:**

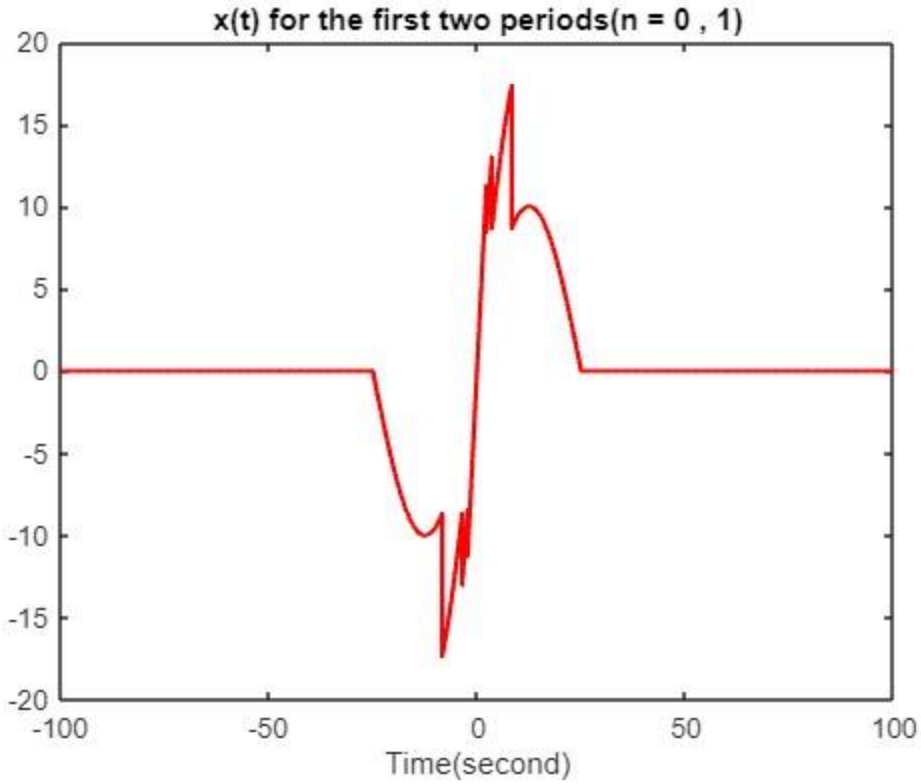
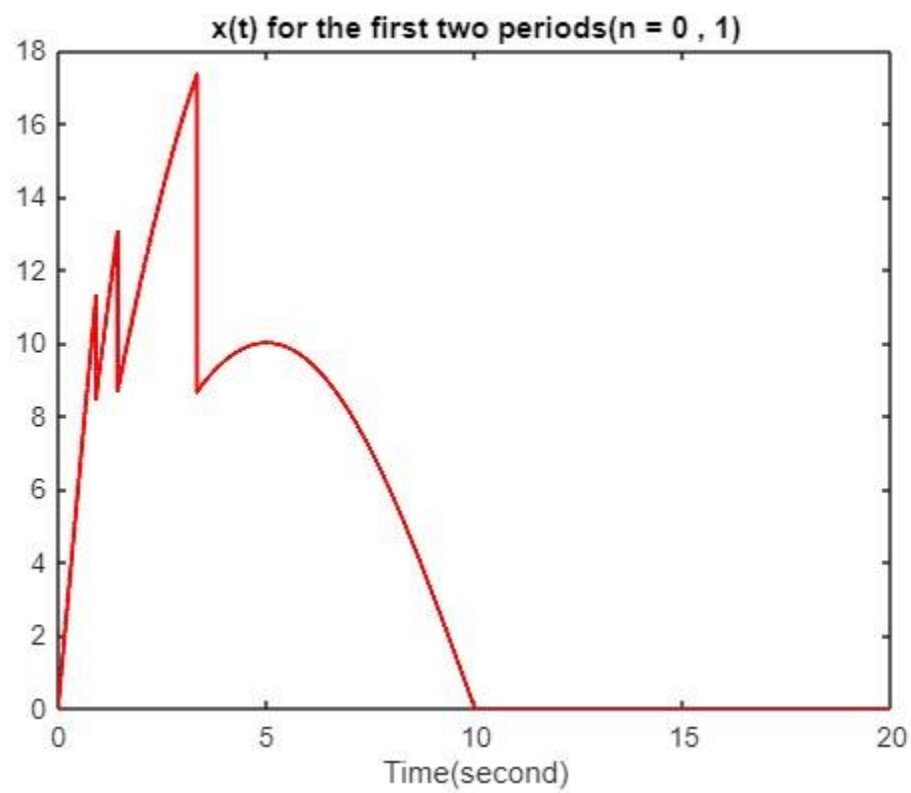
```
T = input('enter the value of T0 :' ) ;  %input command to enter the value of T0
w = (2 * pi) / T ;

TimeInterval = (-2 * T):0.001:(2 * T) ;

x = 10 * sin(w * TimeInterval) .* (rectpuls(((4 * TimeInterval) - (3 * TimeInterval))/T) + rectpuls(((4 * TimeInterval) + (3 * TimeInterval))/T) + rectpuls(((4 * TimeInterval) - (7 * TimeInterval)) / T) + rectpuls(((4 * TimeInterval) + (7 * TimeInterval)) / T));
```

```
plot(TimeInterval , x , 'r' , 'LineWidth' , 1.6) ;
title('x(t) for the first two periods(n = 0 , 1)') ;
xlabel('Time(second)') ;
```

- **Figure:**
  For T0 = 20 ➡



For T0 = 50 ➡



If we changed TimeInterval = 0:0.001:T ; and for T0 = 20